Thomas C. Fraser

Curriculum Vitae

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Personal

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Address	Strynøgade 5, 1.08, 2100 Copenhagen, Denmark
Citizenship	Canadian
Date of Birth	27/11/1994

Positions & education

Jan. 2024 – present	Postdoctoral Researcher Department of Mathematics, University of Copenhagen
Sept. 2018 – Sept. 2023	Ph.D., Physics Thesis: An estimation theoretic approach to quantum realizability problems Perimeter Institute For Theoretical Physics, Waterloo, ON
Aug. 2017 – Jun. 2018	M.Sc., Theoretical Physics, Quantum Information Thesis: A Combinatorial Solution to Causal Compatibility Perimeter Scholars International Program, Perimeter Institute For Theoretical Physics, Waterloo, ON
Sept. 2012 – Apr. 2017	B.Sc., Mathematical Physics, Astrophysics Specialization University of Waterloo, Waterloo, ON

Papers

2026 On the two-copy distillability of Werner states and a new partial trace inequality
Thomas C Fraser, Felix Huber, Balázs Pozsgay, István Vona
arXiv:2607.24309

2026 Complete entanglement detection using polynomial invariants
Thomas C. Fraser, V. Blakaj, R. Rubboli, F. Huber, M. Fanizza
arXiv:2606.16712

2026 On quantum functionals for higher-order tensors
A. Botero, M. Christandl, Thomas C Fraser, I. Leigh, H. Nieuwboer
arXiv:2604.18283

2025 The tetrahedral Horn problem and asymptotics of $U(n)$ $6j$ symbols
A. Alekseev, M. Christandl, Thomas C. Fraser
arXiv:2510.04877

2023 An estimation theoretic approach to quantum realizability problems
Thomas. C. Fraser
Ph.D. Thesis. arXiv:2402.10902

2023 Understanding the interplay of entanglement and nonlocality: motivating and developing a new branch of entanglement theory
D. Schmid, Thomas. C. Fraser, R. Kunjwal, A. B. Sainz, E. Wolfe, R. W. Spekkens
Quantum 7, 1194

2022 A sufficient family of necessary inequalities for the compatibility of quantum marginals
Thomas. C. Fraser
arXiv preprint arXiv:2211.00685

2020 A Combinatorial Solution to Causal Compatibility
Thomas. C. Fraser
Journal of Causal Inference 8(1), 22-53

2018 Machine Learning Peeling and Loss Modelling of Time-Domain Reflectometry
J. R. Rinehart, J. H. Béjanin, Thomas. C. Fraser, M. Mariantoni
arXiv preprint arXiv:1804.04756

2018 Causal Compatibility Inequalities Admitting of Quantum Violations in the Triangle Scenario
Thomas. C. Fraser, E. Wolfe
Physical Review A, 98(2), 022113

Teaching

Block 3 2025-2026	Introduction to Representation Theory - University of Copenhagen Lecturer
Block 3 2024	Introduction to Representation Theory - University of Copenhagen Teaching Assistant
Winter 2021	Quantum Information - Perimeter Scholars International (PSI) Program Teaching Assistant
Winter 2021	Quantum-classical gaps in causal structures - PSI Winter School Student project co-lead
Fall 2018	Physics 1 - Phys 111 - University of Waterloo Teaching Assistant

Work experience

September 2015 – January 2016	Research & Development Data Scientist Meltwater (formerly Sysomos), Toronto, ON, Canada
September 2014 – May 2015	Video Game Developer Lunarch Studios, Waterloo, ON, Canada
January 2014 – May 2014	Mathematics Tutor Humber College, Toronto, ON, Canada
June 2011 – August 2011	Solar Panel Technician OVG Solar, Inc, Renfew, ON, Canada

Programming

Languages	C, C++, Python, Mathematica, Lua, Matlab, Actionscript, JavaScript